

AMENDMENT NO. 4 AUGUST 2025
TO
IS 17264 : 2022 TEXTILES — POLYESTER INDUSTRIAL YARNS —
SPECIFICATION
(First Revision)

(Page 1, clause 3.10) — Insert the following at the end:

‘3.11 Twisted Yarn — Man-made continuous filaments that have been formed by twisting together two or more filaments are known as twisted yarns.’

[Page 5, Table 3, Note (see also Amendment No. 1)] — Insert the following at the end:

‘5 For polyester twisted industrial yarns, the twist shall be declared by the manufacturer subject to the tolerance of ± 5 % on the declared value when tested as per the method prescribed in IS 832 (Part 1).

6 For SI No. (ii), (iii), (v), (vii), (ix), (x), (xi), (xii), and (xiii), the specified linear density range corresponds to the most common varieties prevalent in the market. However, the manufacturers may declare a linear density value beyond this range also.’

[Page 7, clause 8.1(a)] — Substitute ‘Name and description of the material;’ *for* ‘Name and description of the material (see 5.2);

[Page 7, clause 8.1(b)] — Substitute ‘Designation of the material;’ *for* ‘Designation of the material (see 5.1);’.

(Page 9, Annex A) — Insert the following entry at appropriate place:

<i>IS No.</i>	<i>Title</i>
IS 832 (Part 1) : 2021/ISO 2061 : 2015	Textiles — Determination of twist in yarns: Part 1 Direct counting method (<i>third revision</i>)’.

(Page 13, Annex F) — Substitute the following for the existing:

‘ANNEX F

(Table 3)

**METHOD FOR DETERMINATION OF DRY HOT AIR THERMAL SHRINKAGE OF INDUSTRIAL
YARNS USING AN AUTOMATIC THERMAL SHRINKAGE HOT CHAMBER/OVEN**

F-1 GENERAL

F-1.1 This test method covers the measurement of shrinkage of yarns when exposed in a thermal shrinkage hot chamber or oven. The yarn may be exposed to hot chamber in vertical or horizontal position.

F-1.2 This test method is applicable to linear densities in the range from 33 dtex to 7 000 dtex (30 denier to 6 300 denier).

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F-1.3 Yarns or cords for testing may be taken from yarn packages or from fabric.

F-1.4 For specimen tested in horizontal position, the chamber in which the specimens is heated is open on three sides. An effective draft shield shall be used so that air drafts cannot alter the length of specimen.

F-2 PRINCIPLE

Under specific condition, a test specimen is treated in dry-hot air. Shrinkage is calculated as variation in length, measured before and after treatment.

F-3 TEST SPECIMENS

F-3.1 For IDY yarns, strip at least 25 m from the outside of each package in the laboratory sampling unit.

F-3.2 Inspect the outside of the package after stripping off the yarn. If there is visible damage, continue to strip off units of 25 m and reinspect until there is no visible damage.

F-3.3 For testing the sample in horizontal position, take the specimens and put into the clips of the ring of hot chamber. The specimen length will automatically get adjusted to 500 mm to 600 mm as per requirement, as this is the initial distance between the fixed clamp to the movable pulley. Discard and replace specimens that are visibly damaged.

F-3.4 For testing the sample in a vertical position, the specified number of specimens, each of the required length, are cut from each package.

F-3.5 Five specimen of yarn shall be tested and average value of those 5 readings shall be reported.

F-4 TEST PROCEDURE

F-4.1 Test Procedure for Yarns in Horizontal Position

F-4.1.1 With the draft shield in place, set the heater chamber/oven temperature controller set point to give a specimen temperature of $180\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$. This is the temperature for yarns having normal shrinkage level.

F-4.1.2 Refer Table 8 for test conditions.

F-4.1.3 For yarns having low levels of shrinkage, a temperature setpoint of $190\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$ is used.

F-4.1.4 Install one end of the specimens in the fixed clamp.

F-4.1.5 Bring the other end of the specimens over the pulley.

F-4.1.6 Attach a clip-on mass to the unclamped ends of the yarns and start the test.

NOTE — Use a mass that creates a tension load as specified in Table 9.

F-4.1.7 The draft shield will automatically come into place. The specimens are placed automatically in the centre. of heater chamber channel.

F-4.1.8 At the end of 300 s (unless otherwise specified), read the percent shrinkage as indicated on the PC screen attached to instrument output, to the nearest 0.1 percent.

F-4.1.9 Remove and discard the specimens.

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F-4.1.10 The test time may vary depending on specific requirement of customer and/or mutual agreement between producer and customer. A few specific applications may require test time of 900 s (15 min).

F-4.1.11 The required test temperature is generally in the range 177 °C to 180 °C for normal shrinkage yarn.

F-4.2 Test Procedure for Yarns in Vertical Position

F-4.2.1 Hang the conditioned single yarn sequentially from the clamp at the top of the length-measuring stand. Length-measuring stand shall meet the following requirements:

- a) A stand for measuring specimen length of at least 1 mm gauge length;
- b) A millimetre scale to determine specimen length, accurate to within ± 1 mm;
- c) A steel hook or a clamp, where the top of the hook or the lower side of the clamp (where the specimen hung from) is positioned in line with the zero index of the scale; and
- d) Tensioning weights with hooks or clamps for applying tensile force to specimens, accurate to within ± 10 percent.

NOTE — Length of the specimen may also be measured by digital device, provided that the accuracy of the device is ± 1 mm or better.

Carefully add sufficient weight on the bottom of the yarn, to produce the tension as given in Table 9. This tension shall be reached slowly to prevent any over-tensioning. Maintain the tension for (30 ± 3) s.

Mark yarn at the zero index and at a specified length of the scale. Measure the distance between the two marks to get the straightened length (L_0) to an accuracy of 1 mm. Remove weight and single yarn after measurement. Suspend the single yarn from the holder of the oven. Repeat loading and measuring procedure in sequence, until all the single yarns have been tested.

F-4.2.2 Set the oven temperature controller to give a specimen temperature of $180\text{ °C} \pm 2\text{ °C}$. This is the temperature for yarns having normal shrinkage level.

F-4.2.3 Refer Table 8 for test conditions.

F-4.2.4 For yarns having low levels of shrinkage, a temperature setpoint of $190\text{ °C} \pm 2\text{ °C}$ is used.

F-4.2.5 Place the specimen in the heater chamber/oven.

F-4.2.6 Follow the procedures in **F-4.2.1** to measure the distance between the two marks made before thermal treatment, which is the straightened length (L_s), to an accuracy of 1 mm.

F-5 RESULTS

F-5.1 For yarns tested in horizontal position, the shrinkage value is directly displayed on the screen. The shrinkage is the difference between the initial specimen length and final length expressed as a percentage.

F-5.2 For yarns tested in vertical position, the shrinkage in dry-hot air (HAS) is calculated according to formula given below:

$$HAS = \frac{L_0 - L_s}{L_0} \times 100$$

where

HAS = shrinkage in dry-hot air, in percent;

L_0 = length of the test specimen before treatment, in mm; and

L_s = length of the test specimen after treatment, in mm.

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Table 8 Test Conditions
(Clauses F-4.1.2 and F-4.2.3)

Sl No.	Product Type	Temperature °C	Pre-Tension Load g/den	Time s
(1)	(2)	(3)	(4)	(5)
i)	HT/SHT/LE/HMLS	177 to 180 ± 2 °	0.05	120 to 300
ii)	LS/SLS	190 ± 2 °	0.01	120 to 300

Table 9 Pre-Tension Mass for Single Strand Yarn
(Clauses F-4.1.6 and F-4.2.1)

Sl No.	Dtex	Denier	Pre-Tension Mass, Normal Shrinkage yarn g	Pre-Tension Mass, Low Shrinkage yarn g
(1)	(2)	(3)	(4)	(5)
i)	940	840	42	8.5
ii)	1 100	1 000	50	10
iii)	1 440	1 300	65	13
iv)	1 650	1 500	75	15
v)	2 200	2 000	100	20